

REST Assured API Automation Framework Report

Project Title

API Automation Testing Framework for Pet Store APIs using REST Assured and JUnit 5

Github: [ust-sdet-training/SDET at maniprasad-25jun26](https://github.com/maniprasad-25jun26/ust-sdet-training/SDET)

1. Objective

The objective of this project is to design and implement a reusable API automation testing framework using REST Assured to validate Pet Store APIs.

2. User Story

As a QA Automation Engineer,

I want to automate Pet Store API endpoints

So that

I can verify API behavior, validate response contracts, and ensure data integrity without manual testing.

3. API Details

Base URL

[Swagger UI](#) Base Path

Authentication

x-api-key

Query Parameter

project_id=456382

4. Environment & Setup

Hardware

Processor:

Intel Core i5 or above

RAM:

8 GB

Storage:

Minimum 10 GB

Software

Operating System:

Windows 10 / Windows 11

IDE:

IntelliJ IDEA

Programming Language:

Java 17+

Build Tool:

Maven

Test Framework:

JUnit 5

API Testing:

REST Assured

Schema Validation:

JSON Schema Validator

Version Control:

Git

Repository:

GitHub / GitLab

5. Framework Design

Framework follows layered architecture.

Layers

Configuration Layer

Purpose:

Maintain environment variables and reusable constants.

Files:

- **Config.java**
-

Specification Layer

Purpose:

Centralize request and response configurations.

Files:

- **Specfactory.java**
-

Test Layer

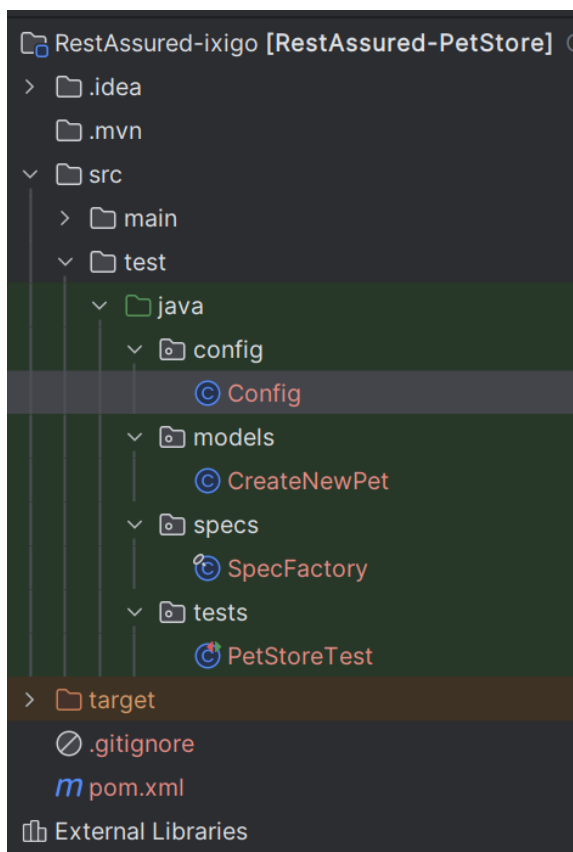
Purpose:

Contains CRUD test implementation.

Files:

- **PetStoreTest.java**
-

6. Project Structure



7. Scenarios Covered

Scenario 1

Create a new pet

Validation:

- **Pet created**
- **Status code**
- **Response body**

Expected:

200 OK

Scenario 2

Retrieve single pet by id

Validation:

- **Pet ID**
- **Pet details**

Expected:

200 OK

Scenario 3

Update existing pet

Validation:

- **Pet updated**
- **Pet name**

Expected:

200 Updated

Scenario 4

Get pet by status

Validation:

- **Retirved pets by status**

Expected:

200 OK

Scenario 5

Delete pet

Validation:

- **error deletion**

Expected:

404

8. Test Execution Flow

Create Pet

↓

get Pet by ID

↓

Update existing pet

↓

Get pet by status

↓

Delete pet

9. How to Run

Open terminal.

Execute:

mvn clean test

Run individual test:

mvn -Dtest=PetStoreTest test

Run from IDE:

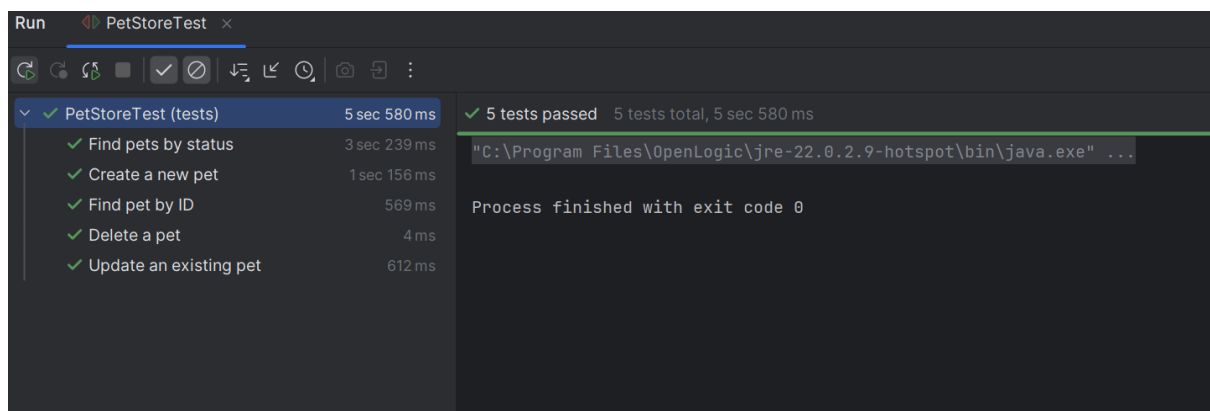
Right Click PetStoreTest → Run

10. Test Results

Scenario	Status
Create Pet	Pass
Get Pet By id	Pass
Update Pet	Pass
Get pet by status	Pass
Delete Pet	Pass

Overall Execution:
Successful

11. Evidence



12. Conclusion

The REST Assured API Automation Framework was successfully developed to automate PetStore APIs.

